



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to NEHRU AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

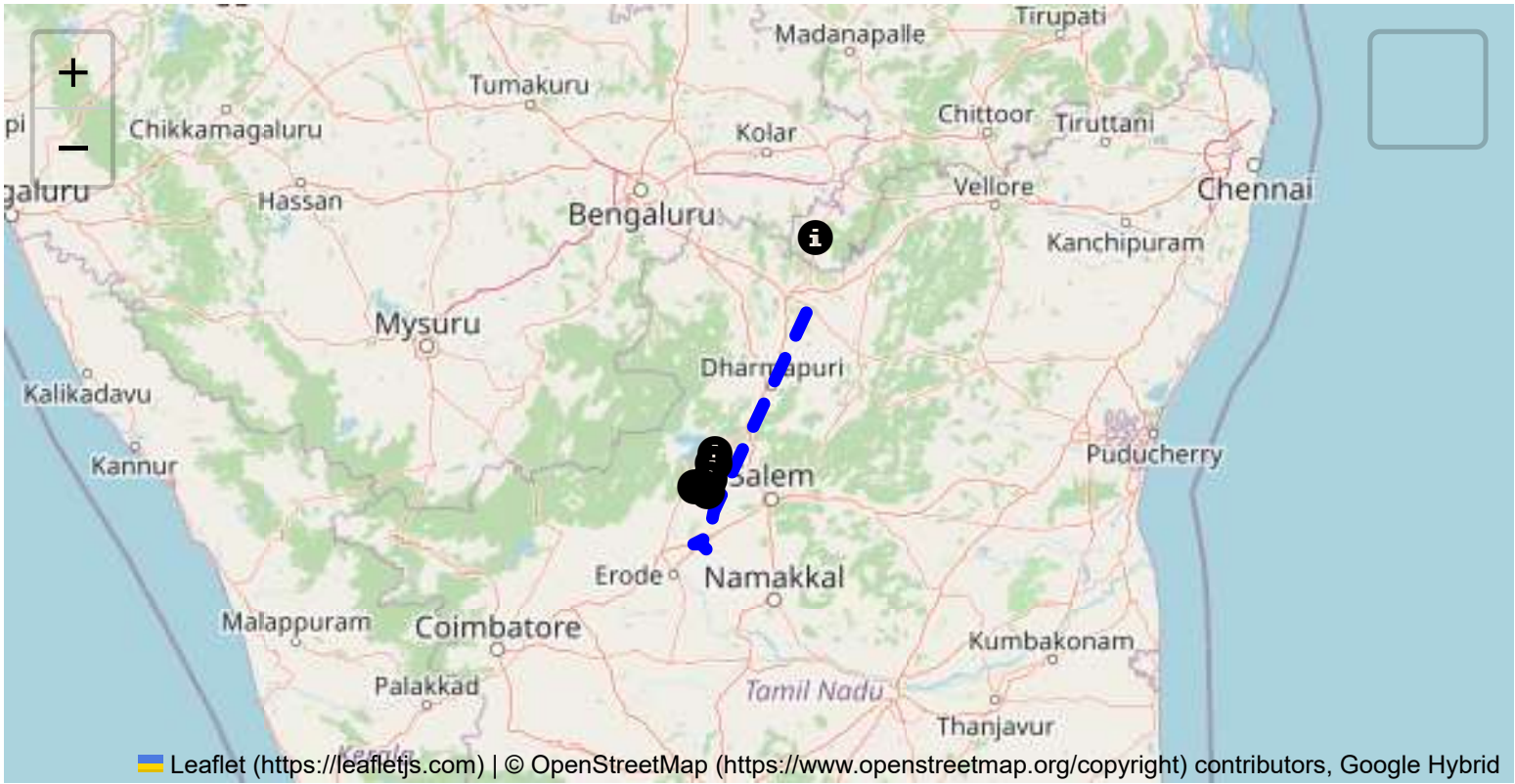
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

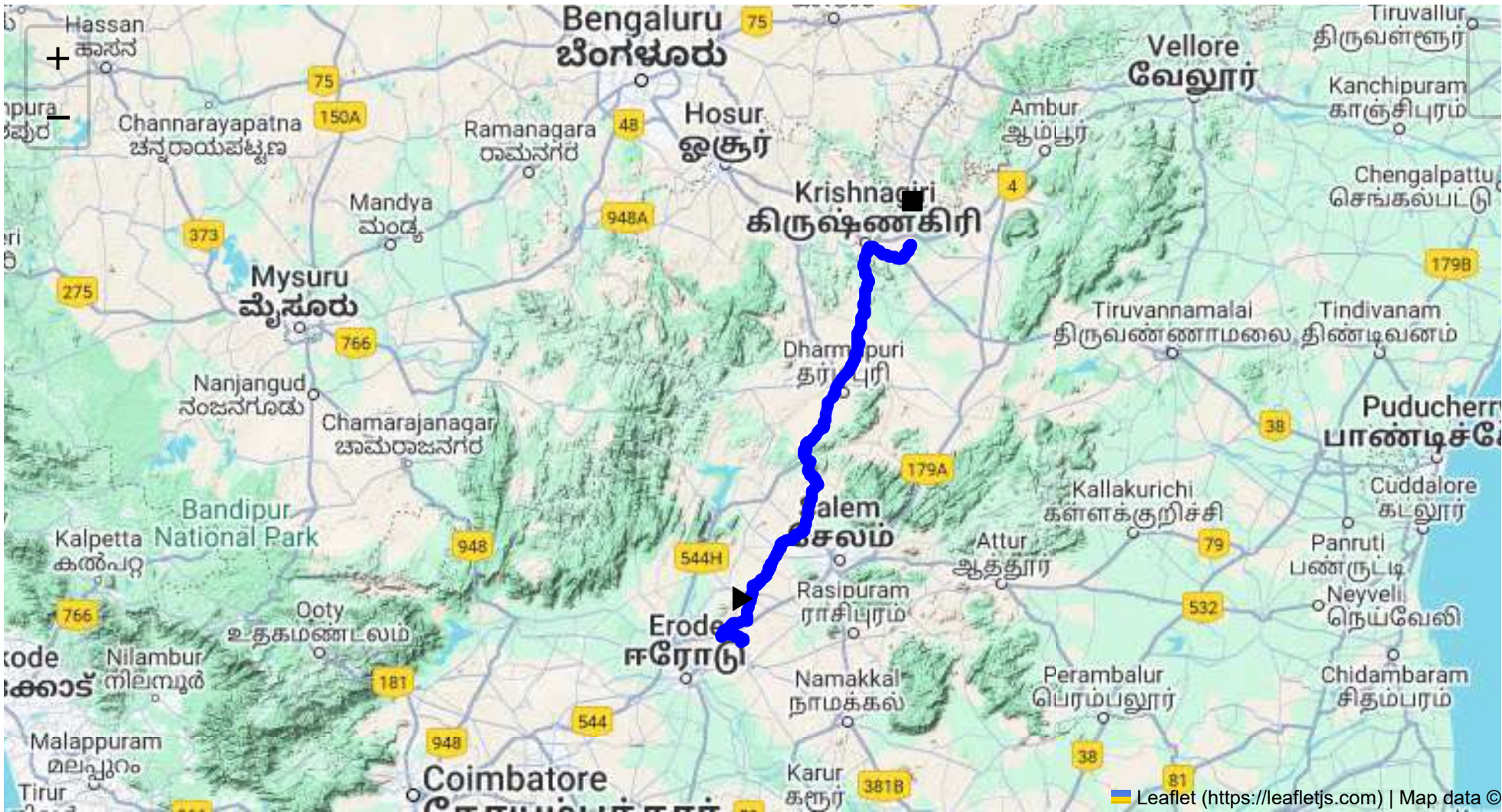
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 167.03 km
Estimated Duration: 3.2 hours
Adjusted Duration (Heavy Vehicle): 4.0 hours
Start: (11.4381, 77.8734)
End: (12.524934, 78.346031)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The specified route begins in Sangagiri, Tamil Nadu (CVQF+23W), and ends in Emakalnatham, Tamil Nadu (G8GW+28Q), covering approximately 167.03 kilometers. This journey follows a path through significant roadways in Tamil Nadu, notably Sankari RS - Erode Privu Road and SH 86, passing through various towns and rural areas en route.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The weather is predominantly hot, with high humidity levels during the summer months. Monsoons arrive from June to September, bringing heavy rain which can lead to flooding and poor road conditions, especially in rural and low-lying areas. Fog can also be a hazard during the early mornings, reducing visibility.

3. Analysis of Traffic Patterns

Traffic tends to peak during the morning (7:00 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM) hours, especially on major roads such as Bhavani Main Road. Congestion-prone areas include market zones and towns like Sankari and Erumaipatti, where local traffic can cause slowdowns.

4. Assessment of Road Quality and Infrastructure

Road quality varies throughout the route. Urban and semi-urban areas have better-maintained roads; however, some stretches, particularly those moving through rural parts, may experience potholes, lack of proper signage, or narrow lanes. The SH 86 is known to have mixed conditions, requiring attentive driving, especially for larger vehicles.

5. Suggestions for Alternative Routes for Emergencies

During emergencies, consider the following alternatives:

- For the segment between Sankari and Erumaipatti, bypass Sankari via smaller district roads connecting through Rasipuram.
- Use NH 544 (Salem-Kanyakumari Highway) for expedited travel to major urban centers like Salem or Erode during a critical situation.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu requires adherence to India's national regulations on hazardous material transport. This includes proper labeling, securing of loads, and compliance with restricted timings for transportation in congested areas to mitigate risks during peak hours. It's essential to secure necessary permits.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While a comprehensive database is limited, anecdotal evidence and local reports indicate that incidents typically involve poor weather and high speeds. There have been cases near urban stretches, where lack of awareness of local traffic laws contributed to accidents involving heavy vehicles.

8. Environmental Considerations and Sensitive Areas

The route passes through a mix of agricultural zones and light forests. It's essential to minimize disruptions to local ecosystems, especially during loads involving hazardous materials. Areas around water bodies and farms are sensitive to spills and emissions.

9. Analysis of Communication Coverage

Mobile communication is mostly reliable along the primary sections of this route, but expect potential dead zones in rural stretches or densely forested areas between Ayveli and Erumaipatti. Ensure that mobile devices are fully charged and consider satellite communication as a backup.

10. Estimated Emergency Response Times for Different Route Segments

Emergency response, particularly medical and firefighting services in Tamil Nadu, is more effective in urban areas with estimated response times of 15-30 minutes. Rural areas may experience delays, with times extending to 45 minutes to an hour, depending on proximity to the nearest town.

11. Overall Summary of Risk Assessment

The route is relatively straightforward yet poses several challenges due to variable road conditions, weather-related issues, and potential communication gaps. It's critical to plan for peak traffic, secure knowledge of local regulations, and maintain open communication channels. Preparing alternative routes and being aware of sensitive environmental areas are essential for safe transportation of hazardous materials.

Comprehensive planning and risk management strategies will ensure smoother operations and heightened safety for truck drivers traversing this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	Medium	11.45654, 77.81706	30 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	High	11.64243, 77.95198	15 KM/Hr
16	Turn	Medium	11.64721, 77.95118	30 KM/Hr
17	Turn	Medium	11.67870, 77.96846	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71416, 77.99604	30 KM/Hr
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	High	11.73775, 78.03465	15 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	High	11.74189, 78.04660	15 KM/Hr
26	Turn	Medium	11.78039, 78.05792	30 KM/Hr
27	Turn	Medium	11.86753, 78.08331	30 KM/Hr
28	Turn	Medium	11.86981, 78.08387	30 KM/Hr
29	Turn	Medium	11.88041, 78.08143	30 KM/Hr
30	Turn	Medium	11.91302, 78.05856	30 KM/Hr
31	Turn	Medium	11.92526, 78.06488	30 KM/Hr
32	Turn	Medium	11.93402, 78.06354	30 KM/Hr
33	Turn	Medium	11.94600, 78.04740	30 KM/Hr
34	Turn	Medium	11.95625, 78.04748	30 KM/Hr
35	Turn	High	11.95880, 78.04989	15 KM/Hr
36	Turn	High	11.96040, 78.04947	15 KM/Hr
37	Turn	High	11.96320, 78.05139	15 KM/Hr
38	Turn	Medium	11.96463, 78.05077	30 KM/Hr
39	Turn	High	11.96597, 78.04920	15 KM/Hr
40	Turn	Medium	11.98504, 78.06287	30 KM/Hr
41	Turn	High	12.24611, 78.19519	15 KM/Hr
42	Turn	Medium	12.25200, 78.20034	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	High	12.40026, 78.21775	15 KM/Hr
44	Turn	Medium	12.40734, 78.22344	30 KM/Hr
45	Turn	Medium	12.50273, 78.22341	30 KM/Hr
46	Turn	High	12.50713, 78.22227	15 KM/Hr
47	Turn	High	12.51545, 78.23078	15 KM/Hr
48	Turn	Medium	12.51540, 78.24086	30 KM/Hr
49	Turn	Medium	12.48218, 78.31195	30 KM/Hr
50	Turn	Medium	12.48382, 78.31736	30 KM/Hr
51	Turn	Medium	12.49393, 78.32753	30 KM/Hr
52	Turn	Medium	12.49710, 78.32879	30 KM/Hr
53	Turn	Medium	12.50128, 78.33423	30 KM/Hr
54	Turn	Medium	12.52154, 78.34403	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
14	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
29	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
32	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
30	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
31	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
33	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45654, 77.81706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71416, 77.99604



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71715, 77.99969



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78039, 78.05792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92526, 78.06488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93402, 78.06354



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.94600, 78.04740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95880, 78.04989



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96463, 78.05077



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96597, 78.04920



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.98504, 78.06287



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.24611, 78.19519



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25200, 78.20034



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40026, 78.21775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40734, 78.22344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50273, 78.22341



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.50713, 78.22227



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.51545, 78.23078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51540, 78.24086



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.48218, 78.31195



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.48382, 78.31736



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.49393, 78.32753



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.49710, 78.32879



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50128, 78.33423



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52154, 78.34403